

Complicated Crown-Root Fracture Treated Using Reattachment Procedure: A Single Visit Technique

Department of Conservative Dentistry and Endodontics, Dental College, Clinical Hospital

Abstract

Trauma to the oral and maxillofacial region occurs frequently, and crown-root fractures comprise a challenging clinical condition. In this report, a single-visit reattachment procedure using a fiber post and composite resin is presented. The patient presented with a fractured maxillary central incisor. A follow-up periapical radiograph demonstrated excellent healing of the periodontal ligament with no radiolucent lesion.

1. Introduction

Crown-root fractures are dental injuries involving enamel, dentin, and cementum, usually extending apical to the gingival margin. Treatment options depend on the fracture line location, biological width invasion, and pulpal exposure.

When the fractured fragment is intact and available, reattachment procedure is considered the most conservative approach. It restores the original tooth morphology, aesthetics, and incisal function while maintaining biological compatibility.

Case Report

A 22-year-old male patient reported to the department following a bicycle fall. Clinical examination revealed a complicated crown-root fracture on the permanent maxillary central incisor with pulpal exposure. The fractured coronal fragment was retrieved in physiological saline.

Figure 1: Intraoral Clinical Photo (Crown Fracture)

Figure 1. Preoperative intraoral photograph showing the complicated crown-root fracture with pulp exposure.

After administration of local anesthesia and rubber dam isolation, single-visit endodontic treatment was executed. A glass fiber post was cemented into the canal, and the tooth fragment was reattached using dual-cure composite resin.

Discussion

Management of subgingival fractures often requires surgical crown lengthening, gingivectomy, or osteotomy. However, when minimal invasion occurs, conservative reattachment provides optimal psychological and functional outcomes for the patient. Histopathology studies demonstrate favorable periodontal response.

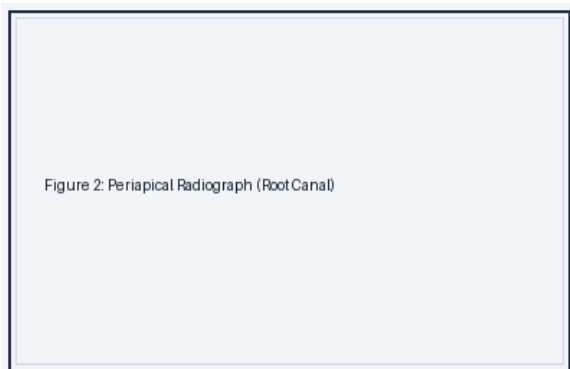


Figure 2. Follow-up periapical radiograph at 12 months showing intact periodontal ligament without radiolucent lesion.

The patient was recalled after 3, 6, and 12 months for clinical evaluation. The restored tooth remained asymptomatic and fully functional.

Conclusion

Reattachment of fractured tooth fragments is an effective, conservative treatment alternative that preserves natural dental tissue and ensures immediate aesthetic recovery.

References

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